

Mc Cluskey Method (With Don't Care)

When, we cannot solve any Boolean expression by K-map (ie. Large no inputs) then we can solve by Mc Cluskey Method.

Find:- Prime Implication and Essential Prime Implementation.

Example: $f = (a, b, c, d) = \sum m (0, 1, 5, 9, 4, 2, 15) + d (13, 6)$

Step 1: Write all number in binary form.

$$0 = 0000$$

$$13 = 1101$$

$$1 = 0001$$

$$6 = 0110$$

$$5 = 0101$$

$$9 = 1001$$

$$4 = 0100$$

$$2 = 0010$$

$$15 = 1111$$

Step 2: Draw table.

Group (No. of 0's/1's)	minterms	Variable	Steps follow
		A B C D	
All 0's / 0	<u>0</u>	0 0 0 0	Find number which have all 0's in given data
No. of 1's, One / 1	<u>1</u>	0 0 0 1	Find number which have only one, 1's in given data
	<u>2</u>	0 0 1 0	
	<u>4</u>	0 1 0 0	
No. of 1's, Two / 2	<u>5</u>	0 1 0 1	Find number which have two, 1's in given data
	<u>6</u>	0 1 1 0	
	<u>9</u>	1 0 0 1	
No. of 1's, Three / 3	<u>13</u>	1 1 0 1	Find number which have three, 1's in given data
No. of 1's, Four (All 1's) / 4	<u>15</u>	1 1 1 1	Find number which have four (all), 1's in given data

Step 3: We match every two adjacent group from top to bottom with each possible bit's pair, if one bit difference (ie. 0 to 1 & 1 to 0) present then it is match pair, otherwise more than one bit difference not considered (ie. Not Match).

Group (If Match)	Matched Pair	Variable	Steps follow
		A B C D	
0	0, 1	0 0 0 X	Match bits between group 0 & 1 for 0 & 1; 0 & 2; 0 & 4 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	0, 2	0 0 X 0	
	0, 4	0 X 0 0	
1	1, 5	0 X 0 1	Match bits between group 1 & 2 for 1 & 5; 1 & 6; 1 & 9; 2 & 5; 2 & 6; 2 & 9; 4 & 5; 4 & 6; 4 & 9 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	1, 6	Not Match	
	1, 9	X 0 0 1	
	2, 5	Not Match	
	2, 6	0 X 1 0	
	2, 9	Not Match	
	4, 5	0 1 0 X	
	4, 6	0 1 X 0	
	4, 9	Not Match	
2	5, 13	X 1 0 1	Match bits between group 2 & 3 for 5 & 13; 6 & 13; 9 & 13 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	6, 13	Not Match	
	9, 13	1 X 0 1	
3	13, 15	1 1 X 1	Match bits between group 3 & 4 for 13 & 15 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).

Note:- Those minterm match then mark as underline in previous table and those minterm not match then left same but remember for final step. (exp.- minterm 5 not match so remember for last and remaining all minterm matched, so mark underline in previous table).

After Simplify:

Group (If Match)	Matched Pair	Variable	Steps follow
		A B C D	
0	<u>0, 1</u>	0 0 0 X	Match bits between group 0 & 1 for 0 & 1; 0 & 2; 0 & 4 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	<u>0, 2</u>	0 0 X 0	
	<u>0, 4</u>	0 X 0 0	
1	<u>1, 5</u>	0 X 0 1	Match bits between group 1 & 2 for 1 & 5; 1 & 6; 1 & 9; 2 & 5; 2 & 6; 2 & 9; 4 & 5; 4 & 6; 4 & 9 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	<u>1, 9</u>	X 0 0 1	
	2, 6	0 X 1 0	
	<u>4, 5</u>	0 1 0 X	
	<u>4, 6</u>	0 1 X 0	
2	<u>5, 13</u>	X 1 0 1	Match bits between group 2 & 3 for 5 & 13; 6 & 13; 9 & 13 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	<u>9, 13</u>	1 X 0 1	
3	13, 15	1 1 X 1	Match bits between group 3 & 4 for 13 & 15 minterms and if any bit change then write "x" in that position for only single bit change. If two bits change then not consider (ie. Not Match).

Step 4: Follow Step 3 again up to pairing of 4 as per 4 inputs.

Group (If Match)	Matched Pair	Variable	Steps follow
		A B C D	
0	0, 1, 1, 5	Not Match	Match bits between group 0 & 1 for 0, 1 & 1, 5; 0, 1 & 1, 9; 0, 1 & 2, 6; 0, 1 & 4, 5; 0, 1 & 4, 6; 0, 2 & 1, 5; 0, 2 & 1, 9; 0, 2 & 2, 6; 0, 2 & 4, 5; 0, 2 & 4, 6; 0, 4 & 1, 5; 0, 4 & 1, 9; 0, 4 & 2, 6; 0, 4 & 4, 5; 0, 1 & 4, 6 minterms and if any bit change then write “x” in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	0, 1, 1, 9	Not Match	
	0, 1, 2, 6	Not Match	
	0, 1, 4, 5	0 X 0 X	
	0, 1, 4, 6	Not Match	
	0, 2, 1, 5	Not Match	
	0, 2, 1, 9	Not Match	
	0, 2, 2, 6	Not Match	
	0, 2, 4, 5	Not Match	
	0, 2, 4, 6	0 X X 0	
	0, 4, 1, 5	0 X 0 X	
	0, 4, 1, 9	Not Match	
	0, 4, 2, 6	0 X X 0	
	0, 4, 4, 5	Not Match	
	0, 4, 4, 6	Not Match	
1	1, 5, 5, 13	Not Match	Match bits between group 1 & 2 for 1, 5 & 5, 13; 1, 5 & 9, 13; 1, 9 & 5, 13; 1, 9 & 9, 13; 2, 6 & 5, 13; 2, 6 & 9, 13; 4, 5 & 5, 13; 4, 5 & 9, 13; 4, 6 & 5, 13; 4, 6 & 9, 13 minterms and if any bit change then write “x” in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	1, 5, 9, 13	X X 0 1	
	1, 9, 5, 13	X X 0 1	
	1, 9, 9, 13	Not Match	
	2, 6, 5, 13	Not Match	
	2, 6, 9, 13	Not Match	
	4, 5, 5, 13	Not Match	
	4, 5, 9, 13	Not Match	
	4, 6, 5, 13	Not Match	
	4, 6, 9, 13	Not Match	
2	5, 13, 13, 15	Not Match	Match bits between group 2 & 3 for 5, 13 & 13, 15; 9, 13 & 13, 15 minterms and if any bit change then write “x” in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	9, 13, 13, 15	Not Match	

Note:- Those matched pair, match then mark as underline in previous table and those matched pair not match then left same but remember for final step. (exp.- matched pair 0, 8 not match so remember for last and remaining all matched pair, matched, so mark underline in previous table).

After Simplify:

Group (If Match)	Matched Pair	Variable	Steps follow
		A B C D	
0	0, 1, 4, 5	0 X 0 X	Match bits between group 0 & 1 for 0, 1 & 1, 5; 0, 1 & 1, 9; 0, 1 & 2, 6; 0, 1 & 4, 5; 0, 1 & 4, 6; 0, 2 & 1, 5; 0, 2 & 1, 9; 0, 2 & 2, 6; 0, 2 & 4, 5; 0, 2 & 4, 6; 0, 4 & 1, 5; 0, 4 & 1, 9; 0, 4 & 2, 6; 0, 4 & 4, 5; 0, 1 & 4, 6 minterms and if any bit change then write “x” in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	0, 2, 4, 6	0 X X 0	
	0, 4, 1, 5	0 X 0 X	
	0, 4, 2, 6	0 X X 0	
1	1, 5, 9, 13	X X 0 1	Match bits between group 1 & 2 for 1, 5 & 5, 13; 1, 5 & 9, 13; 1, 9 & 5, 13; 1, 9 & 9, 13; 2, 6 & 5, 13; 2, 6 & 9, 13; 4, 5 & 5, 13; 4, 5 & 9, 13; 4, 6 & 5, 13; 4, 6 & 9, 13 minterms and if any bit change then write “x” in that position for only single bit change. If two bits change then not consider (ie. Not Match).
	1, 9, 5, 13	X X 0 1	

Step 5: Prepare final table for Prime Implication with matched pairs and unmatched pairs from all steps.

Group (If Match)	Matched Pair	Variable				Steps follow
		A	B	C	D	
0	0, 1, 4, 5	0	X	0	X	Function = $A'.C'$; $A'.D'$ (Prime Implication ie. Maximum group of 1's present in K-map)
	0, 2, 4, 6	0	X	X	0	
	0, 4, 1, 5	0	X	0	X	
	0, 4, 2, 6	0	X	X	0	
1	1, 5, 9, 13	X	X	0	1	Function = $C'.D$ (Prime Implication ie. Maximum group of 1's present in K-map)
	1, 9, 5, 13	X	X	0	1	
Unmatched Pair	2, 6	0	X	1	0	Function = $A'.C.D'$, $A.B.D$ (Prime Implication ie. Maximum group of 1's present in K-map)
	13, 15	1	1	X	1	

Step Final: Prepare final table for Essential Implication with matched pairs and unmatched pairs from all steps.

[illegible]

Note: From previous table variable column find final function using, if '1' then write A/B/C/D and if '0' then write A'/B'/C'/D' and if 'X' then no meaning for define function.

Using K-Map: Check answer is right.

A B/C D	A'.B'	A'.B	AB	A.B'
C'.D'	1	1		
C'.D	1	1	X	1
CD			1	
C.D'	1	X		
Function	$C'.D + A'.D' + A.B.D$			